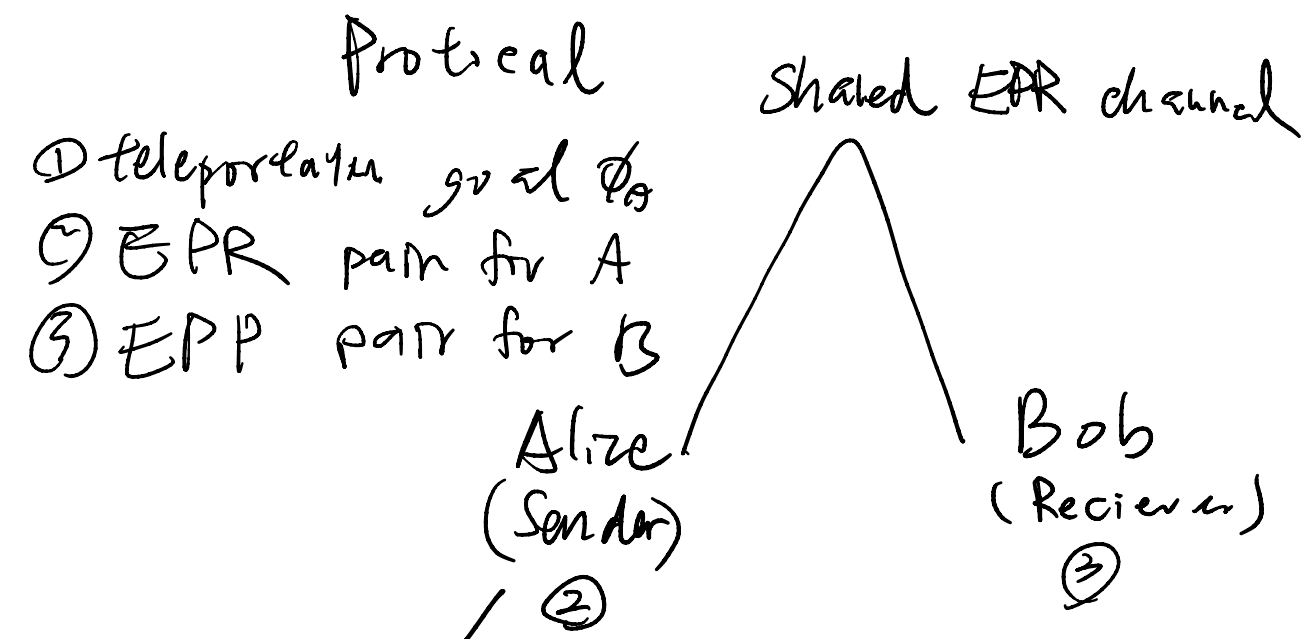


FIG. 1. Spacetime diagrams for (a) quantum teleportation, and (b) 4-way coding [12]. As usual, time increases from bottom to top. The solid lines represent a classical pair of bits, the dashed lines an EPR pair of particles (which may be of different types), and the wavy line a quantum particle in an unknown state $|\phi\rangle$. Alice (A) performs a quantum measurement, and Bob (B) a unitary operation.



- ④
- Step 1: Alice measures ① & ②
- Step 2: Alice sends Bob ① & ② measurement results in the Bell basis
- Step 3: Bob recovers EPR

Resource  EPR states

- Quantum teleportation -

- PPT structure
- ① General theory of QM teleportation
 - ② Experimental results & progress
 - ③ Real-life application

Experimental realization of 1993 paper $\rightarrow$ Uses UV light sources, with BBO crystal.

- ① Uses 2 photon systems
- ② Uses K-photon entanglement as EPR
- ③

BBO crystal generate entangle states using spontaneous parametric down converting type II

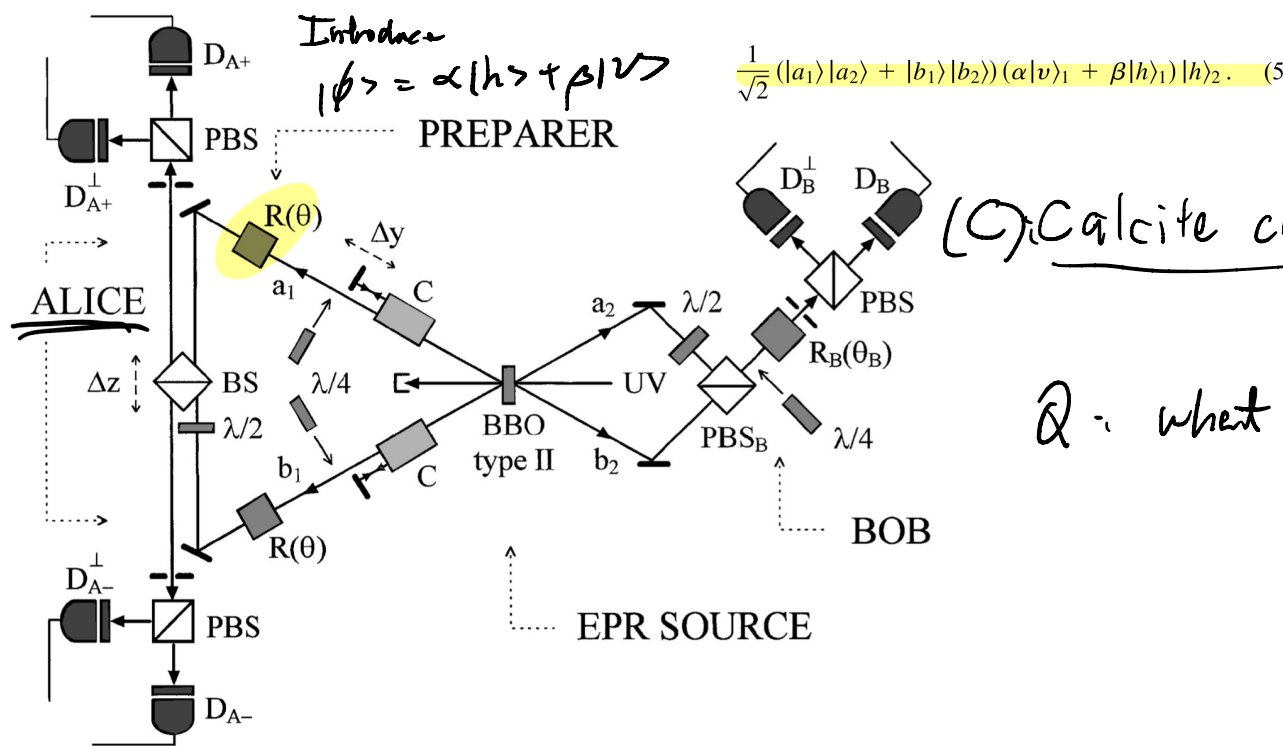
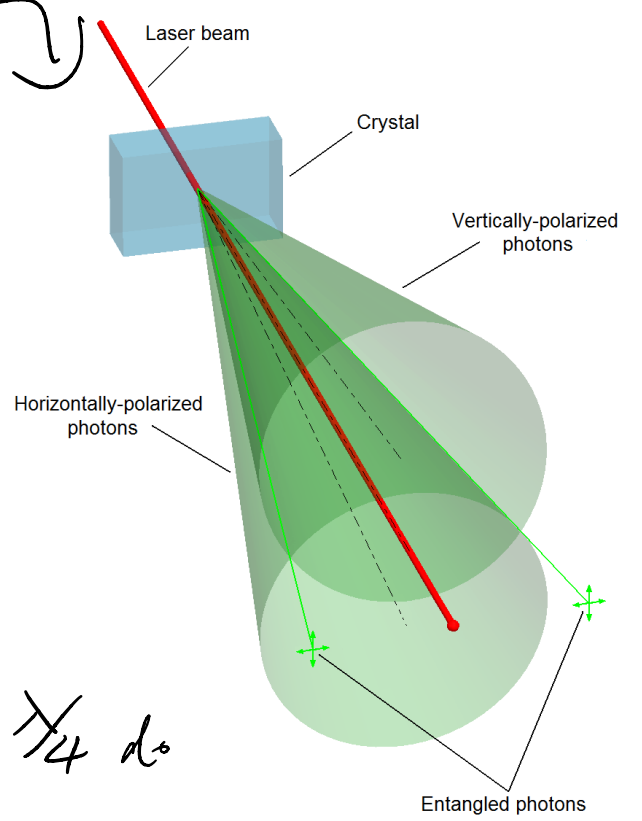


FIG. 1. Diagram of experimental setup showing the separate roles of the preparer, Alice, and Bob.

Non-cloning theorem

Don't exist unitary U for $\forall$ normalized ϕ_A & ϕ_B s.t. $U|\phi_A\rangle|\phi_B\rangle = e^{i\alpha}|\phi_A\rangle|\phi_B\rangle$

(C) Calcite crystal

Q: what does the $\lambda/4$ do